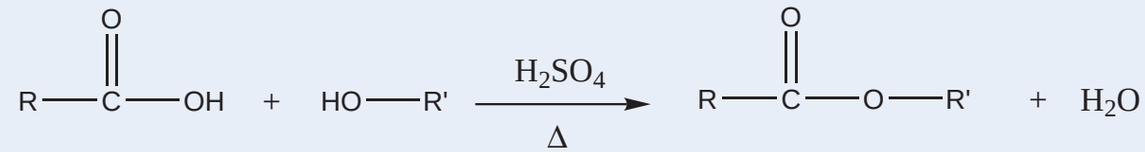
10. Esterification test:

Alcohols react with carboxylic acids to form fruity smelling compounds called esters. This esterification is catalysed by an acid such as concentrated sulphuric acid.



Carboxylic acid — Alcohol — Ester (pleasent fruity odour)

11. Tollen's reagent test:

Aldehydes react with Tollen's reagent to form elemental silver, accumulated onto the inner surface of the test tube. Thus silver mirror is produced on the inner walls of the test tube.

$$\underset{\text{Aldehyde}}{RCHO} + \underset{\text{Fehling's solution}}{2Cu^{2+} + 5OH^-} \longrightarrow \underset{\substack{\text{(Cuprous oxide)}\\ \text{(Red colour)}}}{Cu_2O\downarrow} + RCOO^- + 3H_2O$$

Tollen's reagent preparation:

Tollen's reagent is ammoniacal silver nitrate. It is prepared as follows. About 1 g of silver nitrate crystals are dissolved in distilled water in a clean dry test tube. To this aqueous solution of silver nitrate, add 2 ml of dilute NaOH solution to it. A brown precipitate of silver oxide is formed. This precipitate is dissolved by adding dilute ammonia solution drop wise.

12. Fehling's Test

Fehling's solution A is an aqueous solution of copper sulphate.

Fehling's solution B is a clear solution of sodium potassium tartrate (Rochelle salt) and strong alkali (NaOH).

The Fehling's solution is obtained by mixing equal volumes of both Fehling's solution A and Fehling's solution B that has a deep blue colour. In Fehling's solution, copper (II) ions form a complex with tartrate ions in alkali. Aldehydes reduces the Cu(II) ions in the Fehling's solution to red precipitate of cuprous oxide(copper (I) oxide).

$$\underset{\text{Aldehyde}}{R\text{-}CHO} + 2\underset{\text{Tollen'sreagent}}{\left[Ag(NH_3)_2\right]OH} \longrightarrow \underset{\substack{\text{Metallic silver}\\ \text{(Silver mirror)}}}{2Ag\downarrow} + R-COONH_4 + H_2O + 3NH_3$$

Note: Benzaldehyde may not give this test as the reaction is very slow.

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